Abivan Vijay

Robotics, embedded hardware and web applications

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PROFILE

14-year-old builder who designs and builds ESP32 hardware, autonomous robots and full-stack web applications, and takes them to national competitions, hackathons and expos. Uses AI coding tools (Claude Code, Codex) to take ideas from prototype to working build quickly.

ACHIEVEMENTS

National Robotics Championship 2026, Delhi: drone soccer and innovation contest — **Upcoming** 20–21 Nov 2026

Vadiva, Velammal International School, Ponneri: winners in robo soccer and robo retrieve; robo race — 23–25 Oct 2026

Robotics for Good Youth Challenge (ITU): semi-finalist, national round, Delhi — 10 Oct 2026

Robotics for Good Youth Challenge: selected for the national round, Senior category — Sep 2026

PEC Hacks 4.0: Consolation Prize, Vault (team lead) — Aug 2026

School Shark Tank-style pitch competition: presented Volt, with all 14 branches competing — Aug 2026

Young Environmental Scientist 2026 (IWMA with IIT Madras): participant, Sewer Gas Detector — Aug 2026

National Robotics Championship: selected with CRASH, after qualifying at the SRM zonals — Jul 2026

School-level expo: Winner, FreshSense (team lead) — Feb 2026

RoboWunder International Competition: participant at the Regionals, FreshSense (team lead) — Jan 2026

SRM University, Ramapuram expo: participant, Smart Guard — Nov 2025

PROJECTS

CRASH: Chennai Road Accident Safety Hub *Team project* — JavaScript, Python

Web app that maps road accidents across Greater Chennai, ranks junctions by a severity-weighted risk score and recommends interventions, with analytics dashboards, Monte Carlo simulation and an AI data assistant.

Robotics for Good: Public-Health Response Robot *Team project* — LEGO SPIKE Prime

Fully autonomous robot that runs 2-minute simulated emergency-response missions: moving samples, closing a quarantine zone, delivering medical kits and colour-sorting patients.

Vault: Cold-Chain Integrity Console *Team lead* — TypeScript, ESP32

Monitors a vaccine shipment's cold chain and records every event to a hash-chained ledger that can be verified afterwards.

FreshSense: Food Spoilage Detection *Team lead, 3 members* — ESP32, Blynk IoT

ESP32 conveyor system that classifies food as FRESH or ROTTEN using gas sensors and a self-calibrated baseline, with remote monitoring through Blynk IoT.

Also built: Sewer Gas Detector (ESP32 wearable for sanitation workers, ESP-NOW), Volt (peer-to-peer solar energy ledger, Open Energy Challenge 2026), Smart Guard (team lead, 3 members), WriteWise AI (Flask + Claude), Landmine Detection Shoe (prototype), RFID Door Lock.

SKILLS

Microcontrollers and robotics: Arduino, ESP32, LEGO SPIKE Prime; servo control

Sensors and modules: gas, climate and RFID sensors; displays, servos

Wireless and IoT: Wi-Fi, ESP-NOW (no-router device links); Blynk IoT cloud dashboards

Languages: JavaScript, TypeScript, Python, Arduino C++, HTML, CSS

Backends and data: Node.js, Express, FastAPI, Flask; MongoDB, Supabase; REST APIs, Claude API, Resend

Web: analytics dashboards and maps; canvas and scroll-driven animation

Techniques: sensor calibration; hash-chained ledgers; Monte Carlo simulation

Tools: Claude Code, Codex; Git, GitHub; Render, Vercel

EDUCATION

Velammal Academy, Nolambur, Chennai: Grade 9, CBSE (age 14) — 2026–27